2023 International Conference on Research Methodologies in Knowledge Management, Artificial Intelligence and Telecommunication Engineering (RMKMATE) | 979-8-3503-0570-8/23/$31.00 ©2023 IEEE | DOI: 10.1109/RMKMATE59243.2023.10369395

# Robust Attack Identification Strategy to Prevent Document Image Forgeries by using Enhanced Learning Methodology

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*Abstract*— **Forgery in images is a long-standing issue that spans back to ancient times. Throughout history, images have been employed as evidence or documentation of events. In the present day, the accessibility of image preprocessing tools and affordable hardware has facilitated the creation of manipulated images with ease, often driven by ulterior motives such as disseminating false information or pursuing personal gains unlawfully. In recent years, the spotlight has turned to Artificial Neural Networks (ANNs), sparking significant interest in the realm of image forgery identification. Nevertheless, prevailing image forgery techniques rooted in ANNs are often specialized, geared towards identifying distinct types of forgeries. Consequently, a pressing need arises for an approach adept at efficiently and accurately uncovering unfamiliar forms of image manipulation. This research fills that gap by presenting a robust deep learning-based system designed specifically to spot fakes while working with double image compression. The model's training hinges upon discerning disparities between an original image and its recompressed iterations. To enhance the model's performance, the Ant Lion Optimization (ALO) technique is employed to fine-tune the features of the input data. Noteworthy for its efficiency, the proposed model stands out as a lightweight solution. Empirical results underscore its prowess, showcasing swifter effectiveness compared to prevailing state-of-the-art technique. Encouragingly, the model achieves an impressive overall validation accuracy of 95.87%, reinforcing its efficacy in identifying image forgeries.**

***Keywords: Image Forgery Detection, Artificial Neural Networks, Ant Lion Optimization (ALO) technique, Document Image Forgeries.***

## I. INTRODUCTION

The widespread availability and low cost of electronic equipment is a result of both technical progress and globalization. This has led to the widespread use of digital cameras. All around us are camera sensors, and we utilize them to amass a vast visual database. Numerous scanned photographs are needed for filing various papers online, and many more are posted on social media every day. Visuals are incredible because they can convey meaning even to the non-literate. Therefore, photographs are crucial in the digital era, both for archiving and disseminating information. Numerous tools are at your disposal for instantaneous image modification. These resources were developed to help people get better results while working with photos [1]. However, some people use their ability to fake photos and spread lies rather than improve their image. The harm done by these phony photographs is often irreparable and therefore poses a serious concern.

Humanity has struggled with forgeries for a very long time. A picture is worth a thousand words, as the old adage goes. As technology has progressed, it has become increasingly

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difficult to believe the evidence that is presented. Images play an increasingly important role in today's media of expression. Better picture editing tools and software have been made possible by technological progress.

The branch of machine learning known as "deep learning" has shown to be extraordinarily effective in identifying and making sense of intricate patterns in data. Researchers and practitioners have created cutting-edge methods to spot and stop document counterfeiting by using the power of deep neural networks. In this introductory piece, we'll look at deep learning-based forgery attacks on document pictures, in which attackers employ these methods to build plausible fakes.

In the field of picture forensics known as digital image forgery, many methods are used to examine photos from a given scenario in order to prove their authenticity and trustworthiness. Image processing and computer vision are only two areas where the fast development of deep learning techniques has had a profound impact [2]. Despite the progress that has been made, new difficulties have arisen, most notably in the areas of document security and forgery detection. The growing expertise of counterfeiters and the possible consequences of using forged papers have made document forging an urgent issue.

These digitally altered photos proliferate rapidly over the web and other media platforms. A hazardous and serious situation for the globe has been heightened by the advent of cutting-edge digital technologies, the widespread availability of low-cost processing machines and tools, and the extensive transmissions over numerous websites like Facebook, Twitter, Telegram, Pinterest, and so on. This is a significant issue and contributes to the deterioration of digital image quality. With this in mind, the primary goal of visual forensics is forgery detection [3]. Since these images are often used as legal evidence (Aadhar card, id card, bank passbook), as well as in other official documents (financial documents, newspapers, parts of medical records, and academic records), it is imperative that they be verified as authentic. Figure 1 displays many processed examples of the photos seen in the figure. In addition, social media applications like WhatsApp, Instagram, and Snapchat have led to the widespread adoption of cloud-based picture sharing and editing services like Picasa and Flickr, which in turn have facilitated the widespread distribution of image graphs throughout the Internet.

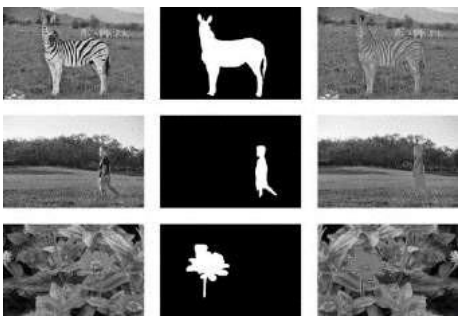


**Fig. 1. A Wide Range of Images and Their Finished Products**

Forgery attacks on document images involve bad actors trying to change or make up the information on official papers including ID cards, passports, bank statements, and contracts [4]. Forgery detection has always required time-consuming and error-prone manual examination by specialists. Automation and efficiency had hitherto been unattainable in this space, but recent deep learning-based technologies have changed all that.

Image forging is used to convey false information by replacing its target's original picture with a fake one. Image fraud has transformed what was once a very reliable medium for sharing information into a tool for spreading falsehoods. The public's faith in pictures is suffering as a result, because it's not always easy to tell if an image has been tampered with [5] [12]. To prevent the spread of misinformation and restore public trust in visual media, it is essential to detect photo forgeries. To do this, a number of image processing techniques may be used to identify forging artifacts. In this study, we introduce a powerful deep learning-based system tailored to the detection of picture forgeries within the framework of double image compression. The key to training our model is learning to recognize differences between the uncompressed and compressed versions of a picture. To enhance the model's feature extraction process, we harness the power of Ant Lion Optimization (ALO).

## II. Related Study

The counterfeiting of currency, academic credentials, artwork, and other papers has increased during the past several years [6]. This is becoming an increasingly critical issue as more and more paperwork must be kept. As a result of individual differences in text characters, pen pressure, writing angle, as well as stroke structures, document forgery as well as forensics are inextricably intertwined fields. As a result, handwriting analysis may be used as a biometric for identity verification and document authenticity checking. However, traditional authorship determination methods are intrusive, time-consuming, and need a high skill level. To that aim, this research investigates hyperspectral imaging's (HSI) viability as a cutting-edge, nondestructive method for uncovering and averting document forgery. Small spectral differences between handwritten samples can be detected [6] because to the granular scene spectral information provided by HSI. This imaging method might be used in many different contexts, from agricultural and ecological surveillance to remote detection as well as forensics to document analysis as well as medical imaging. Convolutional Autoencoder (CAE) for feature extraction, as well as Support Vector Machine (SVM) for author recognition, are the two pillars of our novel unsupervised technique, CAE-SVM. The approach was tested on the UWA writing ink hyperspectral pictures dataset for blue and black inks, and the results were compared to those of state-of-the-art algorithms and convolutional neural network (CNN). In a first, the proposed approach improved accuracy for blue ink to 92.78%. The findings of this research suggest that HSI may be an effective forensic analytical method for spotting and avoiding document forgery [6] [11].

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Image forgery detection [7] may make use of deep learning to assess if an image has been tampered with. This is especially important now, when anybody with a computer and photo editing software may alter images for nefarious or criminal ends. Deep learning algorithms [7] are able to learn to recognize forgery-indicative patterns and traits when trained on large picture datasets. After such algorithms have been developed, we might be able to look at newly taken photographs to detect modification. One way to use deep learning to detect picture forgery is to train a CNN on a database which involves both real and fake pictures. Once the CNN has been taught to recognize particular characteristics, it may be used to verify the authenticity of an unidentified image. Generative adversarial networks (GANs) are another viable alternative. This method [7] uses a combination of two neural networks for image generation and labeling. The discriminator system may be taught to detect the characteristics of forged pictures and accurately classify them by training the GAN on a collection of actual and false photographs. Utilizing deep learning to identify fake photos presents a number of obstacles. These challenges haven't stopped researchers from exploring the potential of deep learning for spotting fake photos. The future dependability of digital images will likely be aided by the application of deep learning algorithms for detecting and avoiding picture forgery [7].

The widespread availability of cameras has contributed to the rise in popularity of photographing events [8]. Images transmit a wealth of information that is vital to our day-to-day lives, thus enhancing photographs to get further information is often necessary. Because of their accessibility, photo editing programs are routinely exploited to create fake photos that distribute false information. This has made the issue of fake artworks far more urgent. Several traditional techniques have evolved throughout time to detect photo forgeries. Recently popularized CNNs have sparked interest in the field of visual forgery detection. However, most CNN-based photo forgery methods in the literature [8] can only detect a single type of fraud. This is why we need a system that can quickly and precisely detect the existence of even the most subtle forms of picture manipulation. In this work [8], we provide a deep learning-based method that is both effective and efficient for identifying photo forgeries under conditions of double image compression. To train our model, we first compare the original image to its compressed counterpart. It has been proven that the proposed model outperforms state-of-the-art techniques while using less computer resources. Positive experimental findings include an overall validation accuracy of 92.23 % [8].

You may find photographs in newspapers, magazines, online, and academic journals [9]. Photoshop, GIMP, and Coral Draw have made it extremely difficult to distinguish between the original and the edited version of a picture. Traditional methods for identifying fake photographs have relied mainly on human-generated attributes. However, the current methods for detecting image manipulation have a major shortcoming in that they can only identify one type of tampering by looking for very specific alterations to the features of the image. Today, deep learning methods are used to detect digitally manipulated images. These methods have shown to be more reliable than traditional ones because of their ability to accurately extract intricate properties from photos. Including analyses, conclusions, and details on publicly available image forgery datasets, this paper [9] provides a complete assessment of deep learning-based systems for photo fraud detection.

Document forgery has been a concern for decades [10] due to the extensive use of paper-based documentation. Traditional destructive procedures cannot be utilized to resolve this issue since they create holes in the document being studied. Hyperspectral imaging is a nondestructive method of identifying an image's unique qualities through spectral fingerprint analysis. Applying deep learning methods to the analysis of hyperspectral images, which are notoriously challenging, has yielded state-of-the-art results. Supervised categorization of hyperspectral pictures is laborious because of the time and effort required to get picture labels and label the training data. In this research, we provide an unsupervised technique for categorizing hyperspectral document images [10].

## III. Methodology

A computer model that mimics the structure and behavior of real neural networks, such those found in the human brain, is called an ANN. It plays a crucial role in the deep learning as well as machine learning processes. ANNs are designed to process and learn from data, enabling them to perform tasks like pattern recognition, classification, regression, and more.

The artificial neuron, sometimes called a node or perceptron, is the network's fundamental element. Neurons take in data, process it, and then send it out as a signal. Typically, neurons in a neural network are arranged in several layers:

i. *Input Layer:* This layer receives the initial data or features that are fed into the network.
ii. *Hidden Layers:* These levels are intermediate among the input as well as output levels. In order to discover useful patterns and characteristics from the incoming data, they execute sophisticated computations. The depth of these layers contributes to the term "deep" in deep learning.
iii. *Output Layer:* The network's ultimate output is generated at this layer, and it might take several forms (such as a categorization label or regression value).

The intensity of the signal sent from one neuron to another is based on the weight assigned to each link between neurons. The neuron also has an associated bias that affects its output. The network's non-linearity is introduced when the weighted sum of inputs and bias is fed into an activation function. The sigmoid, the hyperbolic tangent (tanh), and the rectified linear unit (ReLU) are all examples of activation functions that are commonly used.

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When training a neural network, the weights and biases are adjusted so as to minimize a loss function that measures how far off the actual output is from the expected one. This is typically done through optimization algorithms like gradient descent or its variants. During training, the network iteratively updates its parameters to improve its ability to make accurate forecast on new, unseen data.

Deep neural networks, or systems with several hidden layers, are the subject of deep learning, a subfield of machine learning. With state-of-the-art outcomes in numerous tasks, deep learning has revolutionized many different industries, including image identification, the processing of natural languages, speech recognition, and more.

CNNs are specialized architectures for image and spatial data, while Recurrent Neural Networks (RNNs) are designed for sequence data. There are also more advanced architectures like Transformers, which have proven highly effective in natural language processing tasks.

Overall, artificial neural networks are a foundational concept in modern machine learning and have led to significant advancements in various fields, making them an integral part of the AI landscape.

When one picture is superimposed on top of another, the resulting hybrid will almost certainly have an entirely distinct distribution of DCT parameters than the original. These recurring trends in the histogram are the consequence of the legitimate region being squashed twice: once by the camera and once by the fake. When using the second compression table, the merged chunk performs like a normally compressed chunk would.

As was said before, if a fabricated piece of a photo is present and the picture is recompressed, the fabricated portion will be compressed significantly from the rest of the picture since the origin of the initial picture and the source of the forged portion will be unique. The conceptual diagram of the suggested paradigm is presented in Figure 2. Comparing the altered image to the original reveals a striking degree of fraud when the two are compared side by side. So, we employ it to teach our artificial neural network-based algorithm to spot fake photos.

Create a training set, a validation set, as well as a testing set from your information. Train your ANN using the provided data. Throughout training, the system makes changes to its parameters and weights in order to minimize the loss function's predicted-to-actual label discrepancy. Check your model's progress during training against the validation set to see how well it's doing. Adjust hyperparameters (e.g., learning rate, dropout rate) based on the validation results to achieve the best performance.

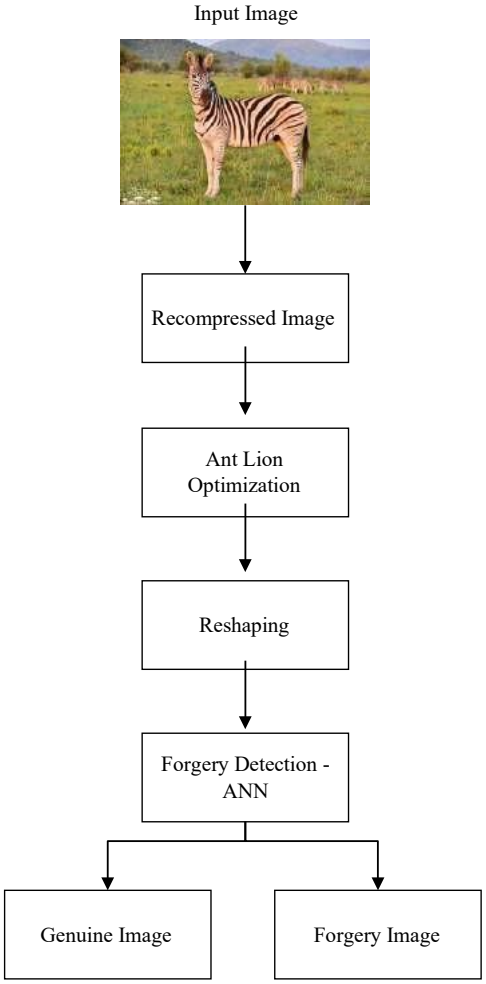



**Fig. 2. Flowchart of the Proposed Method**

In our suggested model, we used a small number of parameters to keep the weight down. The following is a description of the model we built, which consists of three convolutional layers followed by a dense fully connected layer:

- There are 32 3-by-3-inch filters in the first convolutional layer, with a stride size of one and a "relu" activation function.
- There are 32 3-by-3-inch filters in the second convolutional layer, with a stride size of one and a "relu" activation function.
- There are 32 7x7 filters in the third convolutional layer, with a stride size of one and a "relu" activation function, followed by a 2x2 max-pooling layer.
- Then there's the dense layer, which consists of 256 neurons activated by the "relu" function and linked to two neurons activated by the "sigmoid" function (the output neurons).

The network is fed a 128x128 version of the original feature picture. Through the feature photos, the network is able to detect the existence of any manipulation. The suggested algorithm is trained to detect forged images by

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analyzing the wide variety of telltale artifacts that remain after a photo has been tampered with.

### A. Ant Lion Optimization (ALO)

The ALO algorithm is modelled by the catching technique used by ant lions, which involves digging conical trenches to ambush their prey. The program attempts to identify the best possible solutions to optimization issues by modeling their behavior on that of ant lions and ants. Although ALO is often used to optimization problems, it has found usage in other areas, such as forgery detection.

It is important to keep in mind that ALO is not as well-known or as widely used as other optimization algorithms in the machine learning and image processing fields, such as genetic algorithms, particle swarm optimization, or gradient-based optimization techniques. There may not be a well-documented standard strategy to utilize ALO in this context, and as of my last knowledge update in September 2021, its acceptance and implementation in specialized tasks like forgery detection may be restricted.

i. *Problem Formulation:* Define the forgery detection problem as an optimization problem. You would need to map the forgery detection task into a mathematical form that can be solved using an optimization algorithm.

ii. *Fitness Function:* Create a fitness function that quantifies the quality of a solution (or candidate solution) in terms of forgery detection. This function should consider the characteristics of forged and authentic images, and it should guide the optimization algorithm toward solutions that are effective at distinguishing between them.

iii. *Encoding Solutions:* Determine how solutions will be represented in the optimization space. This might involve encoding features, thresholds, or parameters related to your forgery detection algorithm.

iv. *Ant Lion Optimization Implementation:* Implement the ALO algorithm, which typically involves simulating the movement of ant lions and ants within the optimization space. At each iteration, ant lions adjust their positions to find regions with better fitness values.

v. *Local Search and Movement Rules:* Define how ant lions move based on their current positions and fitness values. This movement should aim to improve solutions while also ensuring exploration of the solution space.

vi. *Validation and Evaluation:* Use a validation dataset to evaluate the performance of the solutions generated by ALO. Monitor metrics that reflect the accuracy and effectiveness of forgery detection.

vii. *Parameter Tuning:* ALO has various parameters that can affect its performance. Experiment with different parameter values to find the settings that work best for your forgery detection problem.

The success of ALO its application to forgery detection will depend on various factors, including the complexity of the forgery detection problem, the suitability of ALO for optimization in this domain, and the availability of relevant resources and expertise. The fields of optimization and artificial intelligence are always developing and expanding, therefore it's crucial to keep up with the newest studies and innovations in the area.

## IV. Results and Discussions

In this subsection, we will discuss the context in which the suggested method would be trained and evaluated. In addition, we'll compare its results to those of competing methods.

### a. Experimental Setup

We looked at the suggested method's performance on the widely-used CASIA 2.0 picture forgery database. 7500 are authentic photos and 5100 are fakes out of a total of 12000. CASIA 2.0 has pictures of animals, buildings, articles, characters, plants, nature, sceneries, textures, and interiors, among many other things. Images in the archive come in a wide variety of sizes. The Intel(R) Core (TM) i5-2400 at 3.1 GHz and 16 GB of RAM have been used in the tests.

Initial assessment terms are determined as follows.:

- Total Images: Total number of examined pictures.
- $T_{Pos}$ (true positive): Recognized the altered pictures as real.
- $T_{Neg}$ (true negative): Authenticity of photos was successfully verified.
- $F_{Neg}$ (false negative): False positives are photos that have been checked and found to be fake when they are actually forgeries.
- $F_{Pos}$ (false positive): False positives are original photos that were mistakenly labelled as fakes.

For the purpose of evaluating the suggested technique and making comparisons to others, we compute its accuracy (Acc), precision (Pre), recall (Rec), and F_Score. The formulas for these are as follows:

Here is a formal definition of precision:

$$Acc = \frac{T_{Pos} + T_{Neg}}{Total\ Images} \times 100$$

$$Rec = \frac{T_{Pos}}{T_{Pos} + F_{Neg}}$$

$$Pre = \frac{T_{Pos}}{T_{Pos} + F_{Pos}}$$

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$$F_Score = \frac{2 \times Rec \times Pre}{Rec + Pre} \times 100$$

*b. Model Training and Testing*

In order to test the efficacy of the suggested technique, we randomly split the CASIA 2.0 database into two halves, using 5800 genuine photos and 4000 tampered images (for a grand total of 9800 images) for model training. We utilized Adam as our optimizer, setting the batch size to 64 and the starting learning rate to 1 x 105. To put the suggested model to the test and to compare it to other existing frameworks, we will use the remaining 20% of photos (1500 authentic photographs, 1000 tampered images, total 2500 images). As can be seen in Figure 3, both training as well as testing accuracy dramatically increased when the proposed model was developed on the CASIA 2.0 database with the previously mentioned settings.

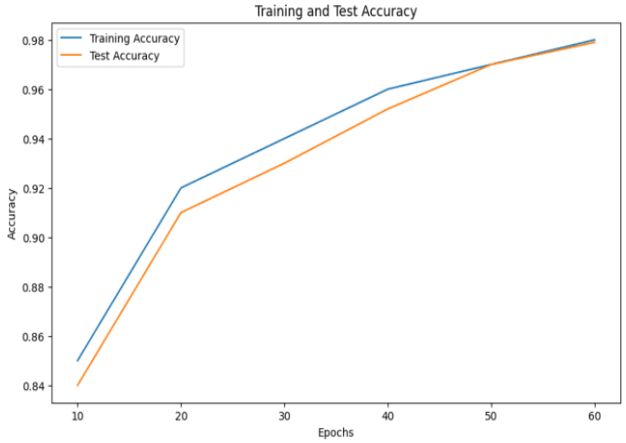



**Fig. 3. Train and Test Accuracy**

*c. Comparison with Other Techniques*

We evaluate how well and how quickly the proposed method detects picture forgeries in comparison to existing methods.

*d. Accuracy Comparison*

The effectiveness of several methods for detecting picture fraud is compared in Table 2. These methods, along with the one being offered, have all been tested on the same CASIA 2.0 database photos. In the case of the aforementioned methods, we classify the input picture as tampered if the mask formed from them indicates forgery, and else we classify it as a genuine image. Many other models have been used as benchmarks for the proposed model. Support Vector Machine, Convolutional Neural Network, AlexNet, MobileNet, and GoogleNet are the preexisting models employed here. Methods are compared and contrasted using the CASIA 2.0 database. We utilized a publicly available, pre-trained model for these techniques in our research. In addition, we updated their models employing the CASIA 2.0 images that were used to create the suggested model. Both the original and retrained versions of these models are included in Table 1. Following retraining, these techniques are more accurate than before, but the recommended technique still excels. Precision, sensitivity, specificity, and accuracy are compared among models in Figures 4, 5, 6, and 7.

TABLE I. PERFORMANCE EVALUATION OF VARIOUS MODELS

| Model | Precision | Sensitivity | Specificity | Accuracy |
|---|---|---|---|---|
| AlexNet | 90.23 | 91.23 | 89.58 | 91.28 |
| SVM | 90.22 | 91.52 | 90.93 | 91.61 |
| CNN | 91.18 | 92.04 | 91.39 | 92.91 |
| MobileNet | 93.21 | 92.06 | 92.23 | 93.92 |
| GoogleNet | 94.08 | 93.98 | 93.18 | 93.75 |
| Proposed | 96.18 | 95.34 | 95.96 | 95.87 |

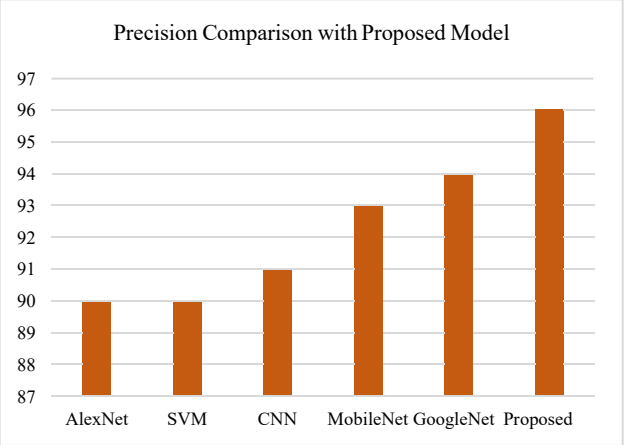



**Fig. 4. Precision Comparison of Various Models**

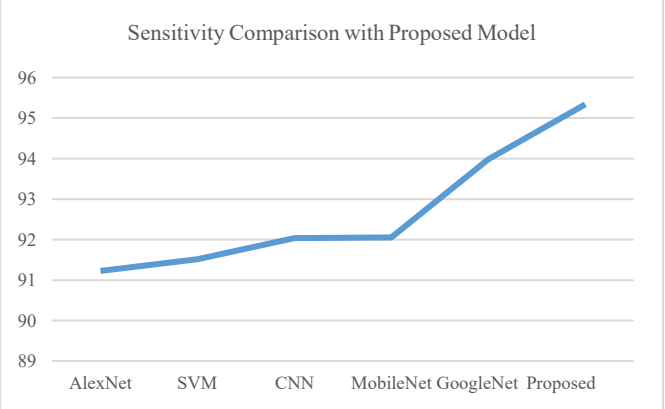



**Fig. 5. Sensitivity Comparison of Various Models**

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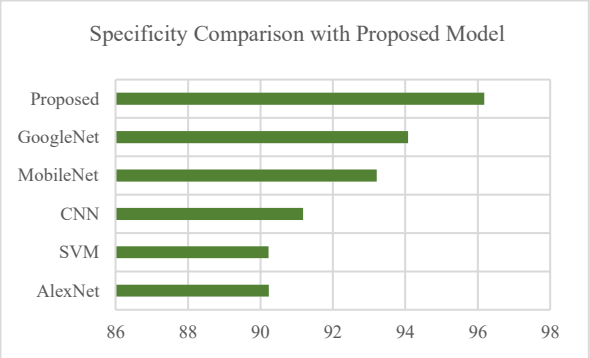



**Fig. 6. Specificity Comparison of Various Models**

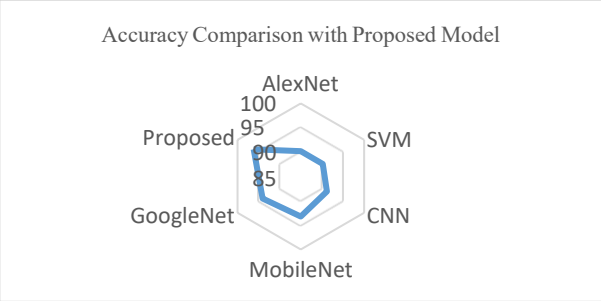



**Fig. 7. Accuracy Comparison of Various Models**

The proposed approach makes advantage of the feature image that is the contrast among the original as well as compressed copies of the picture, to improve the accuracy with which fraud is detected. This aids in detecting image fraud since it is easy to see that the faked component of the picture is emphasized in the feature image. As a result, we have been able to get quite precise results. On the other hand, these methods' low quality in detecting picture fraud stems from the fact that they search for it at the pixel level, where false positive pixels are reported and therefore lower the overall accuracy of the forgery detection at the image level.

We recompressed the image using JPEG compression, as was indicated before; this allows us to choose from a wide range of quality settings. In addition, we have shared the findings of our testing of the proposed model using several different JPEG quality criteria. Accuracy is maximized by maintaining a quality factor of 90 or above. The proposed approach enhances precision by making use of more sophisticated input features, as opposed to systems that merely employ the original image as input characteristics. To achieve this, in addition to the enhanced processed characteristics, we also supply the results of training our algorithm utilizing the raw data. As a result of using the altered input characteristics (the distinction between the uncompressed and compressed versions of the image), the model's accuracy drops from 92.23 to 72.37. When we get a quality factor (JPEG compression) of 98 for the recompressed pictures as well as an accuracy of 95.87% on test images, we may call it a success. Our method outperformed other methods by a wide margin when it came to identifying forged images. Furthermore, our approach can deal with copy-move as well as spliced-image forgeries. However, only a small fraction of state-of-the-art methods are capable of dealing with both forms of picture forgeries, and most can only manage one.

### *e. Processing Time Comparison*

In Table 2, we can see how long it takes for different methods to analyses a picture and make a determination on whether or not it has been tampered with. The suggested method is faster and less expensive than current state-of-the-art methods when it comes to making forgery predictions in images. The mean amount of time required by the suggested approach and the other strategies for identifying forgery in a picture is graphically compared in Figure 8.

TABLE II. COMPUTATION TIME OF VARIOUS MODELS

| Model | Computation Time |
|---|---|
| AlexNet | 4478 |
| SVM | 3486 |
| CNN | 1962 |
| MobileNet | 2286 |
| GoogleNet | 515 |
| Proposed | 57 |

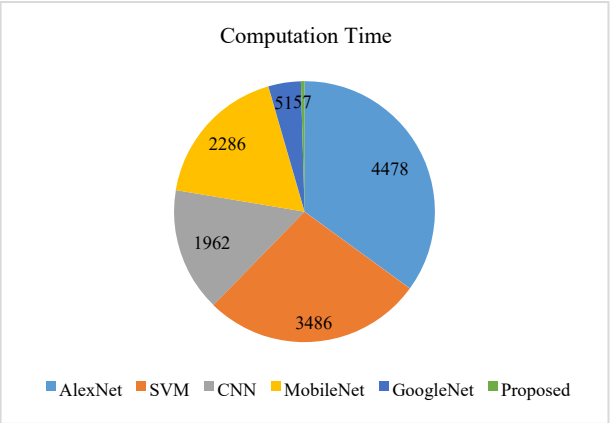



**Fig. 8. Computation Time of Various Models**

The suggested ANN-based model is lightweight compared to existing strategies, and we supply an efficient feature image to our model. This implies that it can make predictions much more quickly than before. This means that our concept has practical applications.

## V. Conclusion and Future Scope

Digital image forgery involves manipulating original photographic images through actions like resizing, rotation, and scaling. When presenting photographic images as evidence in court, it becomes crucial to discern whether a given image is genuine or a forgery. While Artificial Neural Networks (ANNs) have gained prominence in recent years and influenced image forgery detection, many existing ANN-based techniques are tailored to detect specific forgery types. As a result of this shortcoming, there is a pressing need for a

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method that can quickly and reliably detect unsuspected instances of picture alteration. Within the framework of double image compression, we provide a powerful deep learning-based method aimed at detecting picture frauds. Our method makes use of differences between the original and compressed versions of a picture to teach the model. To enhance the effectiveness of the model, we incorporate Ant Lion Optimization (ALO) to fine-tune the input data's features. Notably streamlined, the proposed model stands out for its efficiency. Empirical evidence underscores its superiority, demonstrating swifter effectiveness compared to prevailing state-of-the-art methods. Encouragingly, experimental results reveal an overall validation accuracy of 95.87%, highlighting the model's capability to detect image forgeries effectively.

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